

“THE LUMINESCENT PROPERTIES OF NEON... CONSTITUTE SOURCES OF LIGHT OF GREAT BRILLIANCY.”

George Claude, Vacuum discharge-tube for lighting purposes, US Patent Office, 1915



A gas-discharge light tube works by applying voltage across the low-pressure gas to create ions (charged particles), which causes a colored, neon, light to be emitted.

**IN 1914, BELGIAN PHYSICIAN ALBERT HUSTIN** (1882–1967) found he could **stop blood clotting** by adding a reagent called sodium citrate. Previously, the danger of coagulation meant that transfusions had to be performed directly from donor to recipient. In March, Hustin performed the **first safe nondirect transfusion** using citrate-treated blood from storage.

In 1915, two German scientists would change the way we viewed our world. Geologist **Alfred Wegener** expanded on a **theory of continental drift** that he first proposed in 1912 (see below). Physicist **Albert Einstein's theory of general relativity** was more radical still. His special relativity theory (see 1905) had asserted that measurements of

## ALFRED WEGENER (1880–1930)

Born in Germany, Alfred Wegener began his career as a meteorologist, and he participated in several Arctic expeditions to investigate climate. He is best known for his theory of continental drift. Wegener collected geological evidence but could not explain how the movement occurred; during his lifetime the theory was not taken seriously.



distances and time change for different frames of reference, even though the speed of light stays constant everywhere. The frames of reference referred to situations where the observers

making the measurements were moving at different fixed speeds. Einstein's new theory of general relativity accommodated acceleration and deceleration effects too. Acceleration is

## Blood transfusion kit

The first anticoagulant-treated blood bank kits—used on the frontline during World War I—relied on the knowledge that blood group O could be used as a universal donor.



equivalent to gravity and an outcome was that the effect of gravity was to bend light. Furthermore, he argued that **strong gravitational fields** should distort time and appearance; in other words, they would **warp a space-time continuum**.

Physicists consider space and time to be related: space makes up the three (everyday) dimensions, and time is the fourth dimension.

In the same year, French engineer George Claude received his patent for a new type of lighting: **the neon discharge tube**.

## CONTINENTAL DRIFT

Alfred Wegener collected evidence for his theory that modern landmasses originated from a prehistoric supercontinent that fragmented millions of years ago. He saw not only that transatlantic coastlines seemed to match, but also that geological formations (including fossils) were similar on either side. This suggested that the coastlines were once joined.

